$$H \sim \vec{\sigma} \cdot \vec{B} = \sigma_x B_x + \sigma_y B_y + \sigma_z B_z,$$

where the symbol ~ means "proportional to." Remember that σ_x, σ_y, and σ_z represent the components of the spin *operator* in the above quantum version.

Let's take a simple example in which the magnetic field lies along the z axis. In that case, the Hamiltonian is proportional to σ_z. For convenience, we'll absorb all the numerical constants, including the magnitude of the field (but not $\hbar$), into a single constant ω and write

$$\mathbf{H} = \frac{\hbar\omega}{2}\sigma_z. \qquad (4.23)$$

The reason for the 2 in the denominator will become clear soon.

Our goal is to find out how the expectation value of the spin varies with time—in other words, to determine $\langle\sigma_x(t)\rangle$, $\langle\sigma_y(t)\rangle$, and $\langle\sigma_z(t)\rangle$. To do this, we just go back to Eq. 4.19, and plug in these components of **L**. We get

$$\begin{aligned}
\langle\dot{\sigma_x}\rangle &= -\frac{i}{\hbar}\langle[\sigma_x, \mathbf{H}]\rangle \\
\langle\dot{\sigma_y}\rangle &= -\frac{i}{\hbar}\langle[\sigma_y, \mathbf{H}]\rangle \\
\langle\dot{\sigma_z}\rangle &= -\frac{i}{\hbar}\langle[\sigma_z, \mathbf{H}]\rangle. \qquad (4.24)
\end{aligned}$$

Plugging in $\mathbf{H} = \frac{\hbar\omega}{2}\sigma_z$ from Eq. 4.23, we get